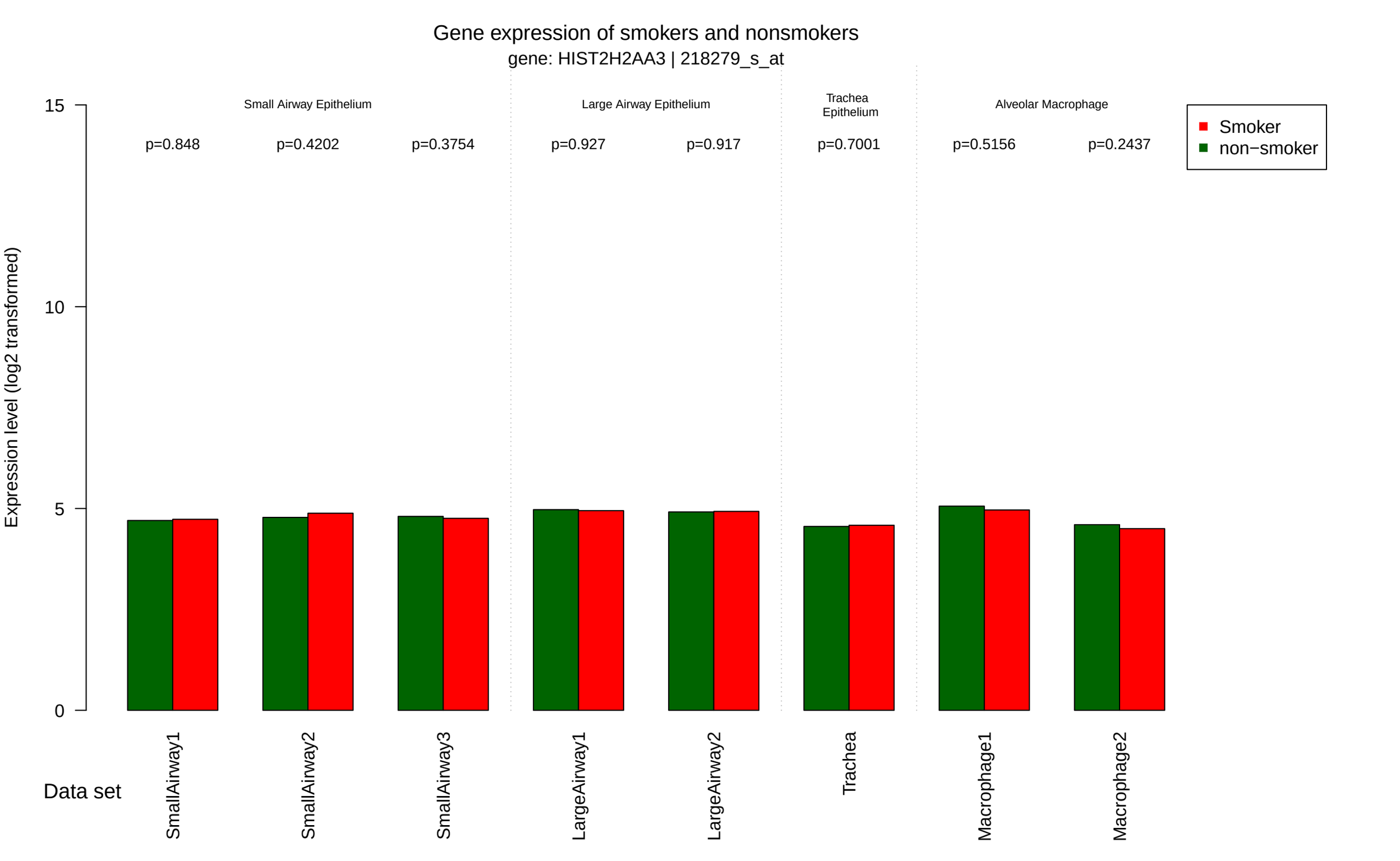
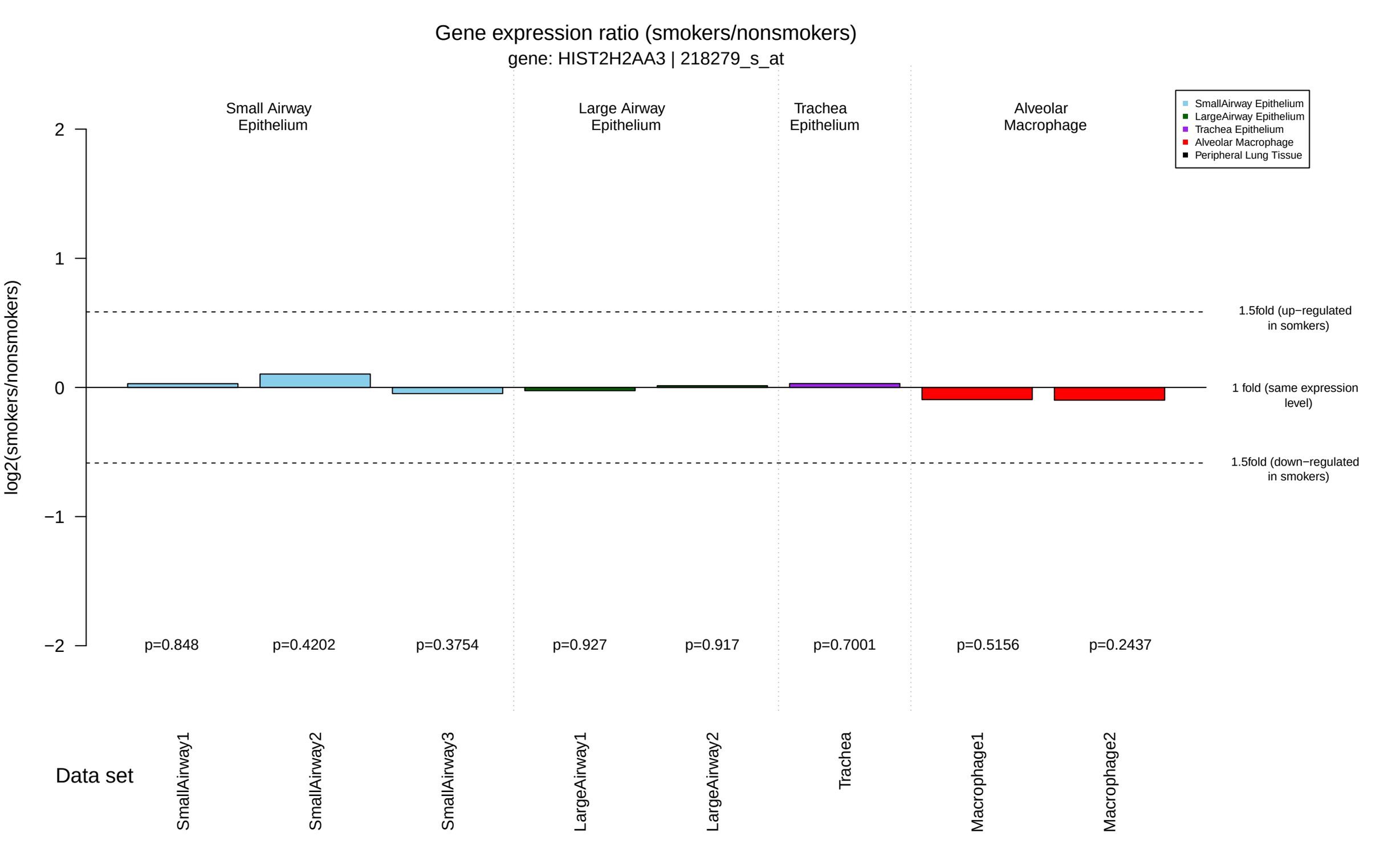
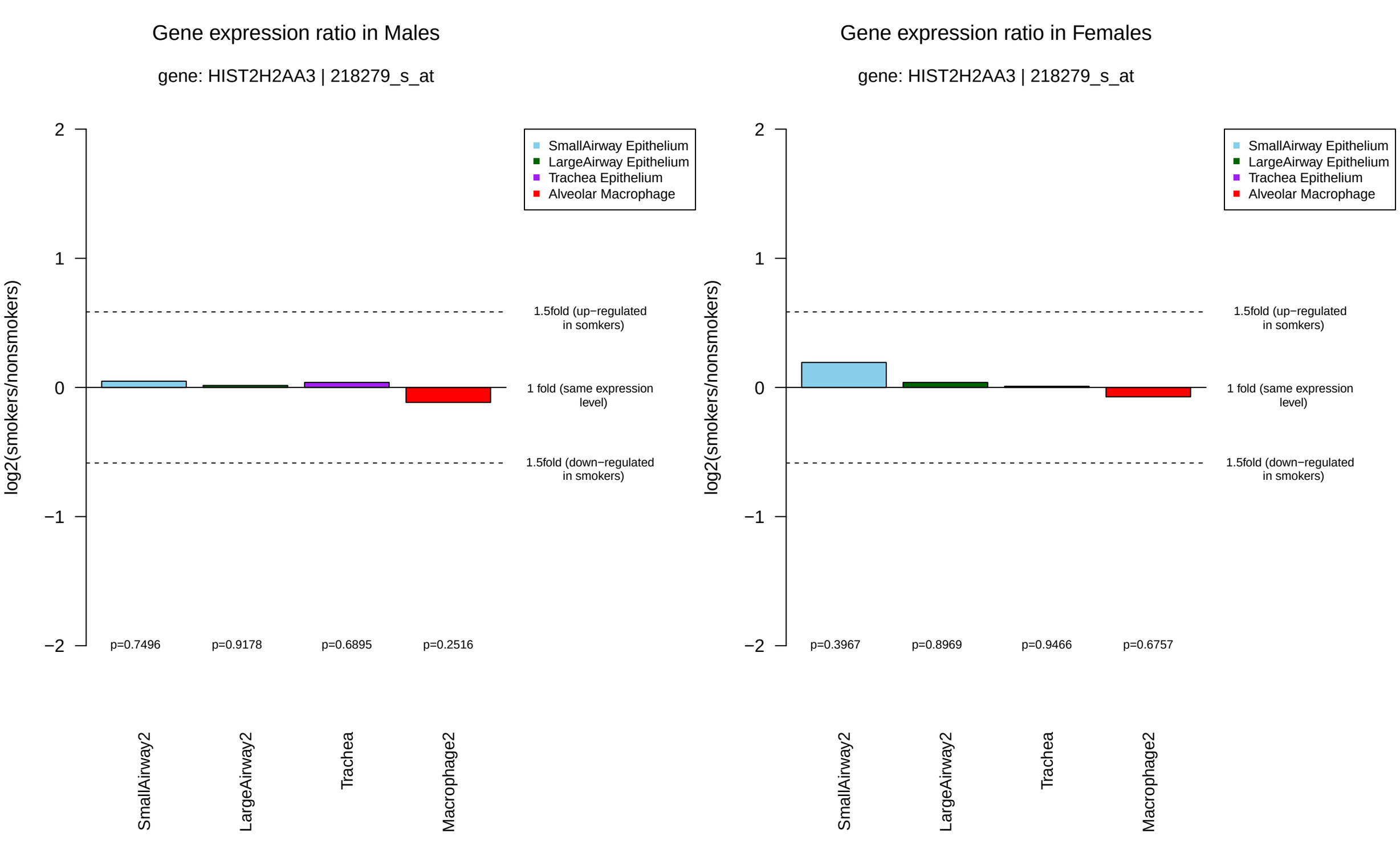
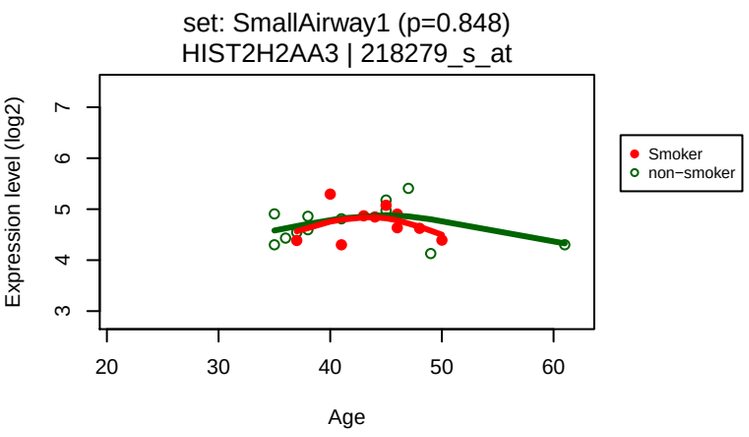
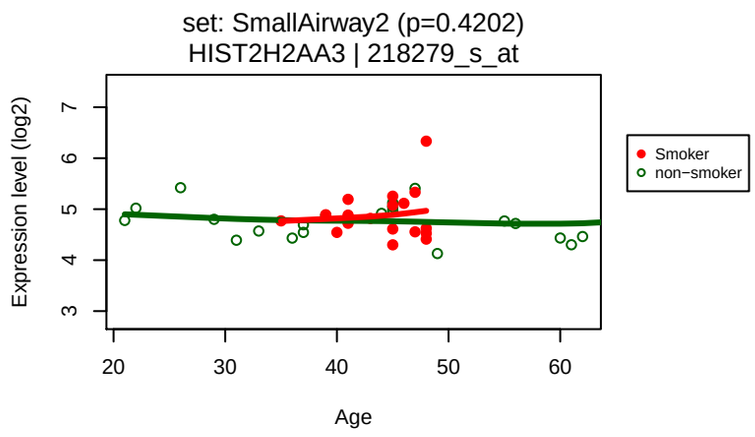
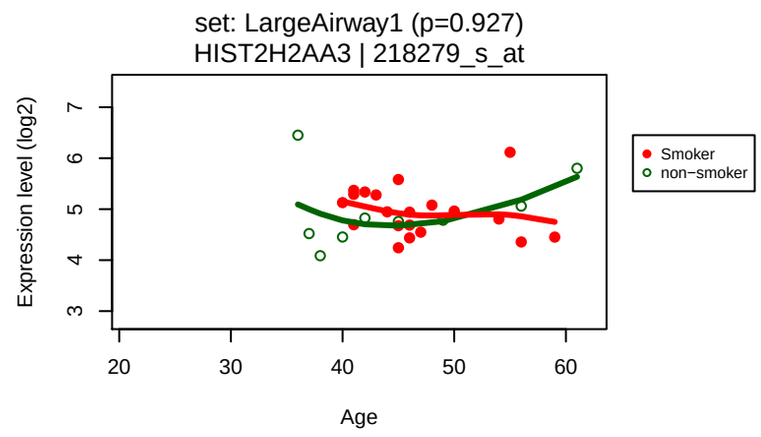

Gene expression of smokers and nonsmokers
gene: HIST2H2AA3 | 218279_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.848
p=0.4202
p=0.3754
p=0.927
p=0.917
p=0.7001
p=0.5156
p=0.2437
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

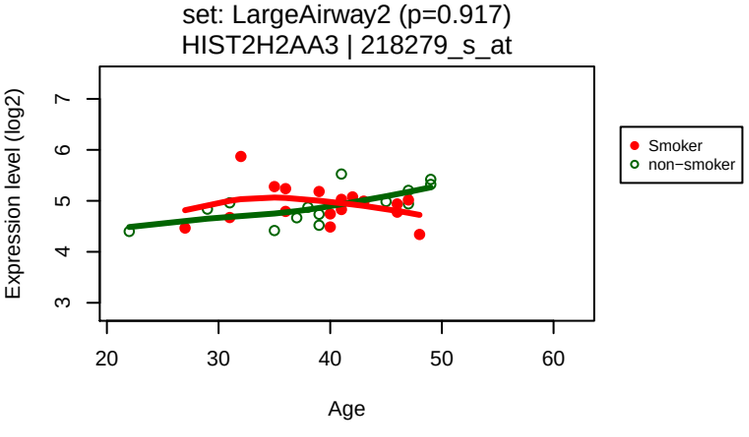

Gene expression ratio (smokers/nonsmokers)
gene: HIST2H2AA3 | 218279_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.848
p=0.4202
p=0.3754
p=0.927
p=0.917
p=0.7001
p=0.5156
p=0.2437
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

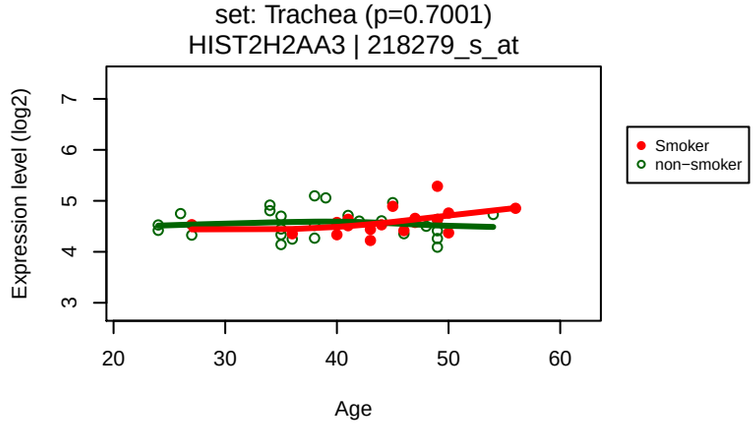

Gene expression ratio in Males
gene: HIST2H2AA3 | 218279_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7496
p=0.9178
p=0.6895
p=0.2516
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIST2H2AA3 | 218279_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3967
p=0.8969
p=0.9466
p=0.6757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

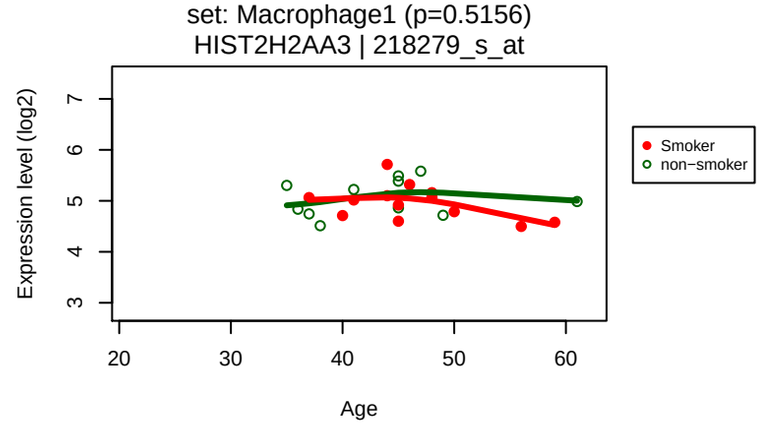

set: SmallAirway1 (p=0.848)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker

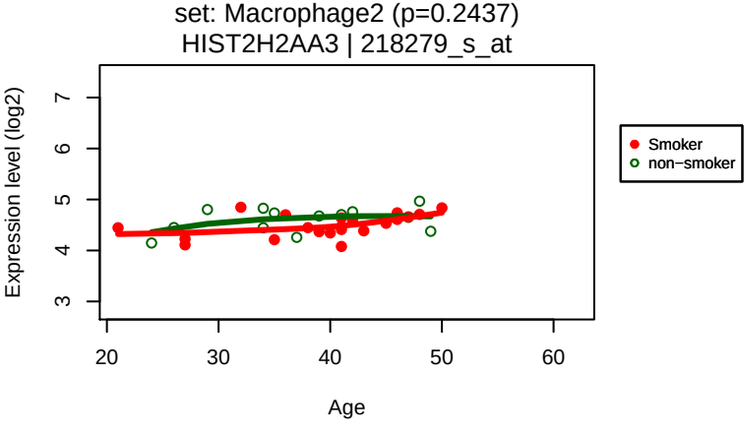

set: SmallAirway2 (p=0.4202)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker

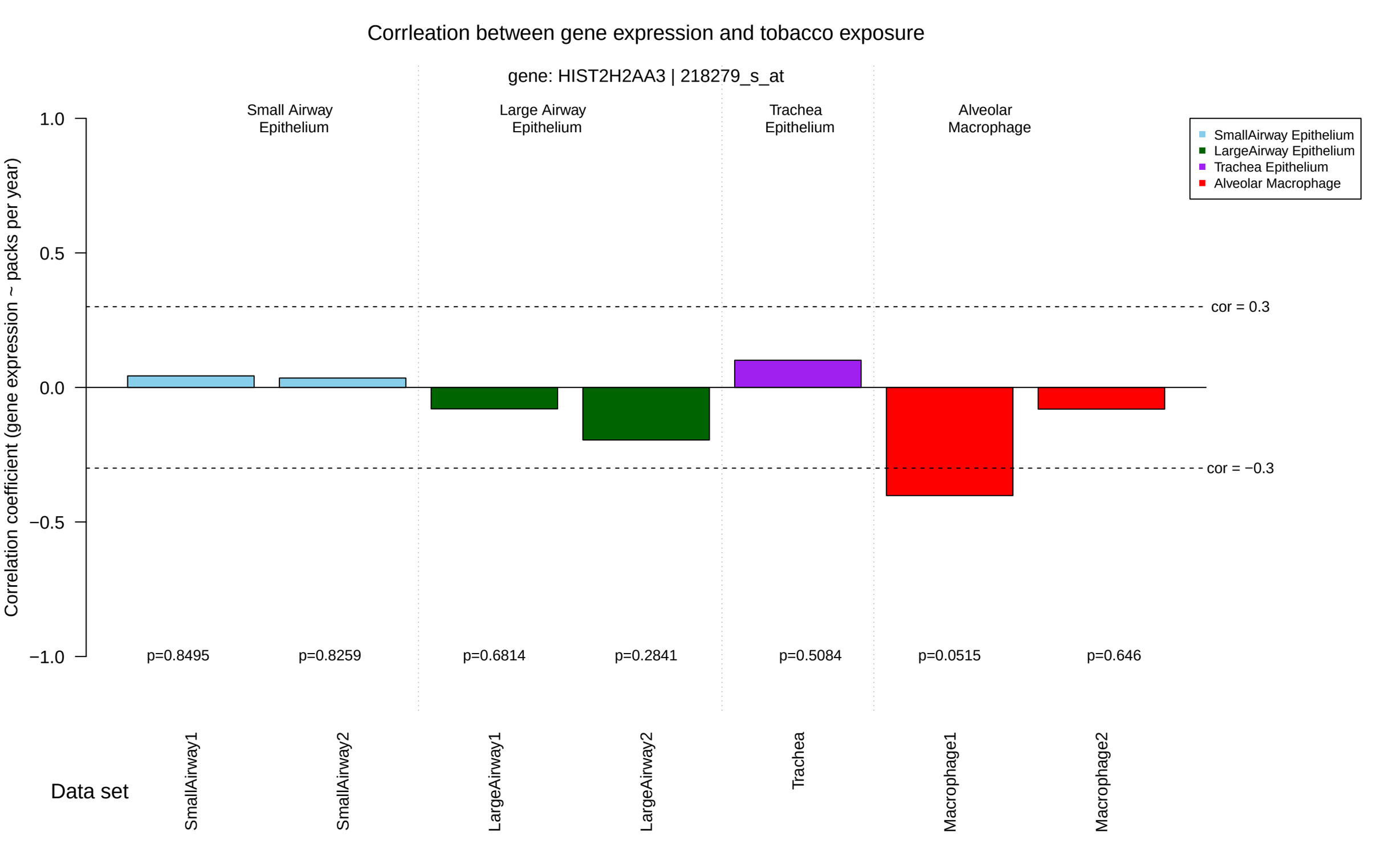

set: LargeAirway1 (p=0.927)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.917)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7001)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5156)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2437)
HIST2H2AA3 | 218279_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIST2H2AA3 | 218279_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8495
p=0.8259
p=0.6814
p=0.2841
p=0.5084
p=0.0515
p=0.646
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2